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Risk-averse dealers in a risk-free market - The role of trading desk risk limits

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1 Big picture

- **Question.** Why did U.S. Treasury dealers reduce risk-taking and liquidity provision during stress in a market whose securities are conventionally viewed as risk-free?
- **Central answer.** Dealers are constrained by self-imposed trading-desk value-at-risk (VaR) limits. These limits make even risk-neutral traders behave as if they are risk averse when the desk approaches its internal risk capacity.
- **Main setting.** The paper studies Treasury market-making desks at large banks from January 2015 through March 2023, using confidential Volcker rule supervisory data on desk positions, VaR, VaR limits, profit and loss, and trading activity.
- **Why VaR limits matter.** VaR is highly sensitive to volatility. During stress, a portfolio can become more constrained even if the position size does not grow, because market volatility raises the desk’s measured risk.
- **Core empirical result.** When a desk’s VaR limit usage doubles, the desk reduces its net securities positions by about 3.1 percent the next day, 5.8 percent over one week, and 7.0 percent over two weeks in the baseline specification.
- **Net versus gross positions.** The effects are larger for net positions than gross positions, consistent with VaR being sensitive to unhedged directional risk and not simply to balance-sheet size.
- **March 2020 event study.** Dealers closer to VaR limits sold more Treasuries to the Federal Reserve’s SOMA portfolio, sold at lower prices, and provided less liquidity to clients.
- **Fed intervention result.** Fed purchases acted like an exit valve. By taking risk off dealer balance sheets, SOMA purchases allowed constrained dealers to buy more bonds from clients, including bonds not directly purchased that day by the Fed.
- **Regulation comparison.** The paper finds little evidence that the supplementary leverage ratio (SLR) is the main high-frequency constraint on Treasury trading desks; internal VaR limits appear more directly relevant.
- **Market liquidity result.** An aggregate dealer VaR-constraint index is strongly related to Treasury market illiquidity, including tail illiquidity, even after controlling for interest rate volatility.
- **Policy takeaway.** During stress, interventions that remove duration/risk from dealer books may support Treasury market functioning more effectively than policies that only relax size-based balance-sheet constraints.

2 Data and measurement

2.1 Volcker rule trading-desk data

- The paper uses supervisory data collected under the Volcker rule, especially Form FR VV-1.
- The data are at the trading-desk level and include desk-level risk limits, usage of limits, profit and loss, trading positions, risk measures, and transaction information.
- Treasury desks are identified by searching desk names and descriptions for words such as “Treasury” and “U.S. Government,” and then validating whether the desks’ descriptions indicate Treasury market-making.
- The main sample contains 18 prime Treasury market-making desks affiliated with 10 large banks from January 1, 2015 to March 31, 2023.
- A robustness sample expands the definition to 40 desks whose names/descriptions include Treasury securities, even if their main function is not pure Treasury market-making.

Interpretation. The contribution is that the paper observes internal trading-desk risk constraints directly, instead of inferring dealer capacity from aggregate positions or prices.

2.2 VaR, VaR limits, and limit usage

- VaR measures the potential future loss on a trading desk over a one-day holding period at a high confidence level, typically 99 percent.
- The VaR limit is the desk-specific internal cap on permissible VaR.
- The key state variable is

$$Limit\ Usage_{it} = \frac{VaR_{it}}{VaR\ Limit_{it}}. \quad (1)$$

- A value near 1 means the desk is close to its limit; a value above 1 means the desk is above the limit and likely faces pressure to reduce risk.
- The average Treasury-desk VaR is about \$4.2 million and the average VaR limit is about \$12.2 million; average limit usage is about 0.30.

Interpretation. Limit usage is a direct measure of how much spare risk capacity a Treasury desk has. The denominator is the internal risk budget; the numerator rises with both positions and volatility.

2.3 Trading positions

- The main outcomes are daily changes in trading-desk positions.
- **Net securities position:** absolute difference between long and short cash Treasury securities positions.
- **Gross securities position:** sum of long and short cash Treasury securities positions.
- **Combined positions:** include both cash securities and derivatives.
- Net positions are the closest measure of directional risk; gross positions are closer to intermediation scale and balance-sheet usage.

Interpretation. If VaR limits are the binding constraint, effects should be stronger for net positions than gross positions, because hedged long-short positions consume less VaR than unhedged exposures.

2.4 SLR data and credit-line drawdowns

- SLR distance is measured as the difference between the bank holding company’s actual SLR and its minimum requirement.
- Since SLR is measured on quarterly averages, it may not bind at high frequency in the same way that daily risk limits do.
- The paper also uses credit-line drawdowns from FR 2052a as a high-frequency proxy for balance-sheet shocks that could affect SLR capacity.
- Drawdowns are measured as changes in outstanding corporate revolving credit facilities, scaled by total BHC assets.

Interpretation. These variables allow the authors to compare risk-based internal constraints with size-based regulatory balance-sheet constraints.

2.5 Fed SOMA purchase data

- The event study uses Federal Reserve emergency Treasury purchase operations from March 16 to June 11, 2020.
- These open market operations are multi-price, multiple-security reverse auctions in which primary dealers submit offers to sell Treasuries to the Fed.
- The authors merge Fed operation results with supervisory TRACE transaction data to infer the amount each dealer sold to the SOMA portfolio.
- The data include dealer-day sale probability, log sale amount, sales to SOMA relative to customer sales, and dealer purchases from customers.

Interpretation. The SOMA operations provide an event-study setting in which the Fed directly removes securities from dealer balance sheets. The key question is whether this improves dealers' ability to serve clients.

3 Model and empirical specifications

3.1 Risk aversion versus VaR constraint

A risk-averse trader with CARA utility over next-period wealth W solves a mean-variance problem:

$$\max U = E[W] - \frac{\gamma}{2}\sigma^2(W), \quad (2)$$

where γ is risk aversion and $\sigma^2(W)$ is portfolio wealth variance.

A risk-neutral trader facing a VaR limit solves

$$\max E[W] \quad \text{s.t.} \quad VaR_\alpha(W) \leq \overline{VaR}_\alpha(W). \quad (3)$$

The Lagrangian is

$$\mathcal{L}(W, \lambda) = E[W] + \lambda (\overline{VaR}_\alpha(W) - VaR_\alpha(W)). \quad (4)$$

If wealth is normally distributed, $VaR_\alpha(W) = Z_\alpha \sigma_W$. Then the constraint can be rewritten as a bound on σ_W^2 , and the Lagrangian becomes observationally equivalent to a mean-variance objective.

Interpretation. A risk limit creates a shadow cost of risk. A trader may be risk neutral personally, but when the desk approaches its VaR limit, taking additional risk is costly because it raises the chance of breaching the internal limit.

3.2 Baseline inventory regression

For Treasury market-making desk i , on day t , the baseline specification is

$$\Delta \text{Log}(\text{Position})_{i,t,t+h} = \alpha_i + \alpha_t + \beta \text{Log}(\text{Limit Usage})_{it} + c + \epsilon_{it}, \quad (5)$$

where $h \in \{1, 5, 10\}$ business days and

$$\Delta \text{Log}(\text{Position})_{i,t,t+h} = \text{Log}(\text{Position})_{i,t+h} - \text{Log}(\text{Position})_{i,t}. \quad (6)$$

- α_i are desk fixed effects.
- α_t are time fixed effects.
- The coefficient β measures how proximity to VaR limits affects future position changes.

Interpretation. A negative β means that desks closer to their VaR limits reduce positions over the next days. Because limit usage is measured using the previous day's limit, the specification reduces mechanical simultaneity.

3.3 Intra-firm fixed effects

To isolate variation across desks within the same firm on the same day, the paper estimates the baseline position regression with firm-by-day fixed effects.

Interpretation. If results persist with firm-by-day fixed effects, the effect is not merely a bank-level risk appetite or balance-sheet capacity shock. It is desk-level variation in how close each desk is to its own limit.

3.4 Instrumental-variable design

The IV strategy instruments Treasury desk VaR limit usage with changes in VaR limits at non-Treasury desks within the same firm:

$$\Delta(VaR\ Limit)_{t-2,t-1}^{k,inc} = \frac{\sum_{j \in inc} [VaR\ Limit_{j,t-1} - VaR\ Limit_{j,t-2}]}{\sum_{j \in inc} VaR\ Limit_{j,t-2}}, \quad (7)$$

$$\Delta(VaR\ Limit)_{t-2,t-1}^{k,dec} = \frac{\sum_{j \in dec} [VaR\ Limit_{j,t-1} - VaR\ Limit_{j,t-2}]}{\sum_{j \in dec} VaR\ Limit_{j,t-2}}. \quad (8)$$

Interpretation. Non-Treasury desk limit changes proxy for firm-wide risk appetite shifts but do not directly depend on Treasury-market client demand. This helps address the concern that Treasury position changes drive VaR limit usage rather than the reverse.

3.5 Regulatory constraints and drawdowns

The SLR specification is

$$\Delta \text{Log}(\text{Position})_{i,t,t+h} = \alpha_i + \alpha_t + \beta_1 \text{Log}(\text{Limit Usage})_{it} + \beta_2 \text{SLR dist}_{i,q(t-r)-1} + \beta_{int} \text{Log}(\text{Limit Usage})_{it} \times \text{SLR dist}_{i,q(t-r)-1} + \epsilon_{it}. \quad (9)$$

The drawdown specification adds credit-line drawdowns and their interaction with SLR distance:

$$\begin{aligned} \Delta \text{Log}(\text{Position})_{i,t,t+h} = & \alpha_i + \alpha_t + \beta_1 \text{Log}(\text{Limit Usage})_{it} + \beta_2 \text{Drawdown}_{i,t,t+h} \\ & + \beta_3 \text{SLR dist}_{i,q(t-r)-1} + \beta_4 \text{Drawdown}_{i,t,t+h} \times \text{SLR dist}_{i,q(t-r)-1} + c + \epsilon_{it}. \end{aligned} \quad (10)$$

Interpretation. These tests ask whether risk limits simply proxy for regulatory balance-sheet constraints. The results generally point back to VaR limits as the more direct high-frequency constraint.

3.6 SOMA event-study regressions

The event study relates prior-day limit usage to dealer sales to the Fed, offer aggressiveness, and client liquidity provision. Typical outcomes include:

- probability of selling to SOMA;
- log quantity sold to SOMA;
- sales to SOMA relative to dealer sales or customer buys;
- offer aggressiveness, defined as the accepted auction price minus the dealer's inferred sale price;
- log dealer buy volume from customers.

Interpretation. These regressions identify whether desks closer to their limits used Fed purchases as a way to shed risk and whether that created capacity to serve clients.

3.7 Profitability and liquidity regressions

Dealer profitability is measured as the future change in PnL scaled by VaR:

$$\frac{1}{h} \sum_{k=1}^h \left[\frac{PnL}{VaR} \right]_{i,t+k} - \left[\frac{PnL}{VaR} \right]_{i,t} = \alpha_i + \alpha_t + \beta \text{Log}(\text{Limit Usage})_{it} + c + \epsilon_{it}. \quad (11)$$

The aggregate dealer constraint index is

$$\text{Limit Usage Index}_t = \sum_i w_{it} \cdot \text{Log}(\text{Limit Usage})_{it}, \quad (12)$$

with weights based on recent customer transaction volume:

$$w_{it} = \frac{\sum_{k=1}^{20} \text{Cust Trans Volumn}_{i,t-k}}{\sum_i \left[\sum_{k=1}^{20} \text{Cust Trans Volumn}_{j,t-k} \right]}. \quad (13)$$

The paper then estimates quantile regressions of Treasury market illiquidity:

$$Q(\tau)_{\text{Illiquidity},t} = \alpha + \beta_\tau (\text{Limit Usage Index residual})_t + \gamma (\text{MOVE index})_t + \epsilon_t. \quad (14)$$

Interpretation. If constrained dealers charge higher compensation and the aggregate constraint index predicts illiquidity, then desk-level risk constraints scale up into market-wide liquidity conditions.

4 Identification and empirical design

4.1 What identifies the main effect

- The baseline tests use daily variation in each desk’s VaR limit usage.
- Desk fixed effects absorb persistent differences across desks.
- Time fixed effects absorb common market-wide shocks.
- Firm-by-day fixed effects isolate cross-desk variation within the same bank and day.
- The IV design uses non-Treasury desk limit changes to proxy for exogenous shifts in firm risk appetite.

Interpretation. The strongest evidence is not one coefficient, but the alignment of multiple tests: baseline regressions, intra-firm comparisons, IV estimates, regulatory horse races, and the March 2020 event study.

4.2 Why March 2020 is a useful event study

- March 2020 created a sudden surge in Treasury selling pressure and volatility.
- VaR limit usage spiked because VaR rises mechanically with volatility.
- The Fed then purchased Treasuries at very large scale, allowing dealers to remove positions from their balance sheets.
- The paper compares high- and low-VaR-usage dealers before and after the SOMA operations.

Interpretation. If VaR limits are binding, dealers with tighter limits should be more eager to sell to the Fed and more able to provide client liquidity after selling to the Fed. That is what the paper finds.

4.3 Alternative explanations addressed

- **SLR constraints:** results are robust after controlling for SLR distance and credit-line drawdowns.
- **Balance sheet size only:** effects are larger for net positions than gross positions.
- **Firm-wide shocks:** firm-by-day fixed effects still yield negative coefficients.
- **Risk limit endogeneity:** IV estimates using non-Treasury desk limit changes remain negative and often larger.
- **Measurement differences across firms:** firm sensitivity tests and alternative limit-usage measures support the same conclusion.
- **Other risk limits:** DV01 limits matter, but VaR limit usage remains economically larger and statistically significant.

Interpretation. The paper's identification strategy is designed to show that VaR limits are not merely a label for regulation, market stress, firm-level balance sheet constraints, or interest-rate risk limits.

4.4 What the paper cannot fully prove

- The paper observes internal VaR limits but cannot fully observe every internal approval process or trader compensation incentive.
- Limit usage is not randomly assigned; it reflects positions, volatility, and bank risk-management decisions.
- The March 2020 event study is powerful but occurs during an extraordinary crisis with multiple simultaneous policy actions.
- The paper focuses on large Treasury dealers subject to Volcker reporting, not all market participants.

Interpretation. The paper provides unusually direct evidence on dealer constraints, but the evidence is still observational and institutional rather than a randomized experiment.

5 Tables and figures

5.1 Figures 1-2: Treasury stress and VaR limit usage



Figure 1: Dealer inventory index and 10-year Treasury yield

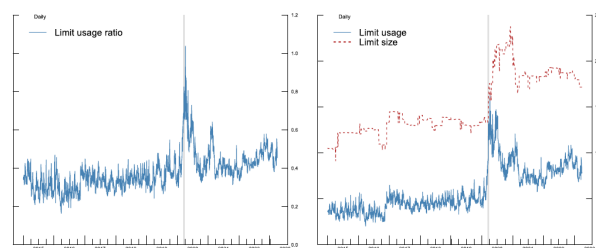


Figure 2: VaR limit size, usage, and usage ratio

Read:

- Figure 1 shows that Treasury dealers reduced inventory during the March 2020 stress while yields moved sharply, consistent with dealers not leaning against selling pressure.
- Figure 2 shows average VaR limit usage rising sharply in March 2020, approaching or exceeding one for some desks.
- The right panel of Figure 2 shows that VaR rose faster than limits: firms raised limits, but not enough to offset the volatility shock.

Economic message. The stress episode looks like a sudden loss of dealer intermediation capacity. The natural mechanism is that volatility raised measured VaR and pushed desks toward internal limits.

5.2 Figure 3 and Table 1: positions, drawdowns, and summary statistics

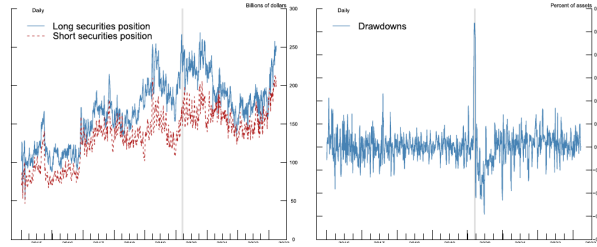


Figure 3: Trading desk positions and bank credit-line drawdowns

Read:

- The left panel of Figure 3 shows long and short securities positions rising over time, with both large levels and substantial time-series variation.
- The right panel shows the spike in corporate credit-line drawdowns around March 2020.
- Table 1 reports an average net securities position of \$4.95 billion and average gross securities position of \$26.7 billion.
- VaR averages \$4.20 million; VaR limits average \$12.2 million; limit usage averages 0.30.
- In the SOMA event-study sample, dealers sell to SOMA on 82 percent of dealer-days, and SOMA sales average 19 percent of total sales to customers.

Economic message. The data are rich at the desk-day level, and the summary statistics show both large positions and meaningful variation in risk capacity.

5.3 Tables 2-3 and Figure 4: baseline position effects

Table 2
Dealer desk positions and VaR constraints (baseline).

(a) Net positions						
	Net securities positions			Net combined positions		
	(1) h = 1	(2) h = 5	(3) h = 10	(4) h = 1	(5) h = 5	(6) h = 10
log(Limit Usage)	-0.031*** (0.011)	-0.058*** (0.014)	-0.070*** (0.016)	-0.046*** (0.017)	-0.091*** (0.021)	-0.116*** (0.023)
Desk FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.129	0.125	0.132	0.131	0.127	0.144
N	16 838	16 786	16 721	12 045	11 986	11 915
(b) Gross positions						
	Gross securities positions			Gross combined positions		
	(1) h = 1	(2) h = 5	(3) h = 10	(4) h = 1	(5) h = 5	(6) h = 10
log(Limit Usage)	-0.007*** (0.002)	-0.020*** (0.003)	-0.030*** (0.003)	-0.006*** (0.001)	-0.017*** (0.002)	-0.024*** (0.003)
Desk FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.132	0.153	0.154	0.136	0.146	0.154
N	16 838	16 786	16 721	12 045	11 986	11 915

Table 2: Baseline: dealer desk positions and VaR constraints

Table 1

Summary statistics.

(a) Positions and VaR						
	mean	sd	p25	p50	p75	count
Net securities position	4.95	9.01	0.23	2.15	6.41	43 310
Gross securities position	26.7	39.2	2.25	13.8	30.0	43 310
Long securities position	15.0	21.4	1.15	7.98	17.8	43 313
Short securities position	11.7	18.9	0.67	4.73	13.2	43 323
Net combined position	350.5	1209.1	0.34	3.24	40.5	31 641
Gross combined position	3020.4	5475.9	10.0	340.7	4097.2	31 641
Long combined position	1676.7	2987.1	5.24	182.4	2145.5	31 657
Short combined position	1344.0	2609.4	4.72	157.1	1740.2	31 686
VaR	4.20	6.25	0.82	2.26	4.73	42 483
VaR limit	12.2	12.1	5	10	15	41 904
Limit Usage	0.30	0.20	0.16	0.26	0.40	41 814
Total PnL	1.01	4.35	-0.088	0.24	1.28	44 217
Total PnL/VaR	-0.35	37.3	-0.049	0.19	0.56	41 950
SLR dist	2.40	2.37	1.30	1.61	2.36	29 863
Credit draws	2.21	2.66	0.66	0.99	2.81	25 935

(b) SOMA event study						
	mean	sd	p25	p50	p75	N
Probability of Selling to SOMA	0.82	0.38	1.00	1.00	1.00	620
Log(Pur Amount of Sale to SOMA)	6.05	2.04	5.00	6.33	7.53	521
Sales to SOMA/ Total Sales to Customer	0.19	0.19	0.02	0.13	0.33	609
Sales to SOMA/ Total Buys from Customer	0.25	0.30	0.03	0.14	0.39	610
Log(Dealer Buy from Customer), SOMA	14.46	1.76	14.00	14.89	15.51	610
Log(Dealer Buy from Customer), not SOMA	14.55	1.54	14.18	15.05	15.51	610
Log (Total Dealer Buy from Customer)	15.34	1.37	14.97	15.81	16.23	610

Table 1 Panel (a) shows summary statistics of Treasury trading desks for the levels of the positions variables (in billions of dollars), VaR and VaR limit (in millions of dollars), Limit Usage (defined as the ratio of VaR relative to VaR limit from the day before). Net positions are the absolute value of the difference between long and short positions. Gross positions are sum of long and short positions. Securities positions include only cash assets. Combined positions include cash and derivative assets. The sample period is January 1, 2015 to March 31, 2023 for securities positions and January 1, 2015 to December 31, 2020 for combined positions. Total profit and loss (PnL) is in millions of dollars, or scaled by VaR. SLR distance is distance in percent between the observed and the required SLR. Credit draws are draws on revolving credit facilities at the BHC level, as percent of total BHC assets. Panel (b) shows summary statistics for data used in the SOMA event study, for the period March 16, 2020 to June 11, 2020. Sales to SOMA and Sales and Buys to and from Customers are in millions of dollars. Log(Dealer Buy from Customer), SOMA (not SOMA) is the log volume of trades that are dealer buy from customers for bonds that are included (not included).

Table 1: Summary statistics

Table 3

Net positions relative to gross positions.

	Net/Gross securities positions			Net/Gross combined positions		
	(1) h = 1	(2) h = 5	(3) h = 10	(4) h = 1	(5) h = 5	(6) h = 10
log(Limit Usage)	-0.023** (0.011)	-0.035** (0.014)	-0.038** (0.015)	-0.038** (0.017)	-0.068*** (0.021)	-0.086*** (0.022)
R ²	0.130	0.125	0.132	0.132	0.128	0.145
N	16 838	16 786	16 721	12 045	11 986	11 915

Table 3 shows the effect of log(Limit Usage) on the log change in the ratio of net securities positions to gross securities positions (columns 1 through 3) and on the log change in the ratio of net combined positions to gross combined positions (columns 4 through 6). Net and gross securities positions include only cash assets. Net and gross combined positions include cash and derivative assets. Net positions are the absolute value of the difference between long and short positions. Gross positions are the sum of long and short positions. Limit Usage is defined as the ratio of VaR relative to VaR limit from the day before. Regression results are shown for positions change over h = 1, 5, and 10 business days.

Table 3: Net positions relative to gross positions

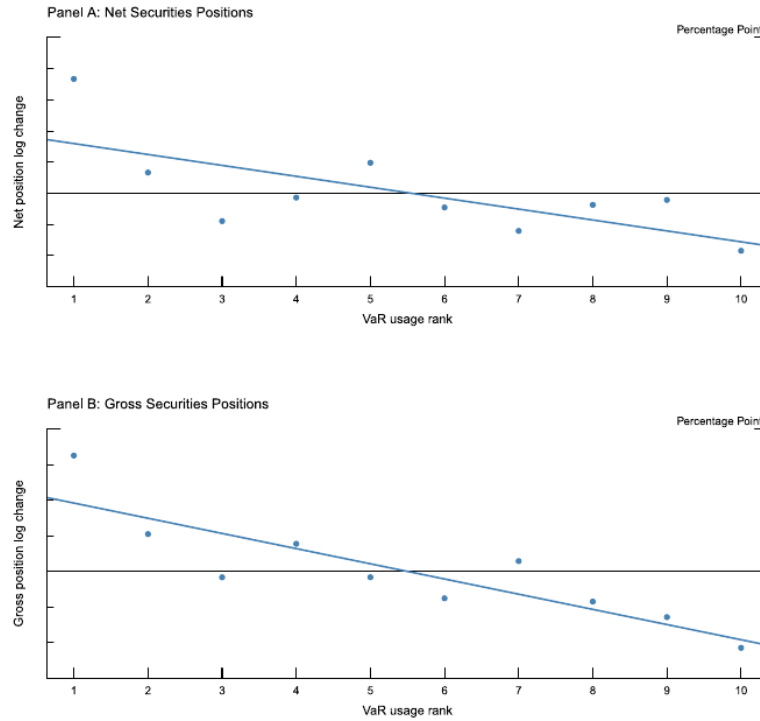


Figure 4: Position changes against VaR limit usage rank

Read:

- Table 2 shows that higher log limit usage predicts reductions in both net and gross positions.
- The net securities position coefficients are -0.031, -0.058, and -0.070 over 1, 5, and 10 business days.
- Combined net positions show even larger negative effects: -0.046, -0.091, and -0.116.
- Gross-position effects are negative but smaller, consistent with VaR limits focusing on risk rather than pure size.
- Table 3 shows that higher limit usage reduces the ratio of net to gross positions, especially for combined positions.
- Figure 4 visualizes the negative relationship between VaR usage rank and position changes.

Economic message. Dealers close to internal VaR limits actively reduce risk exposure. The fact that net positions respond more strongly than gross positions points to risk limits rather than pure balance-sheet limits.

5.4 Tables 4-6: within-firm effects, inventory vintage, and IV analysis

Table 4
Dealer desk positions and VaR constraints with firm-by-day fixed effects.

(a) Net positions						
	Net securities positions			Net combined positions		
	(1) h = 1	(2) h = 5	(3) h = 10	(4) h = 1	(5) h = 5	(6) h = 10
log(Limit Usage)	-0.028 (0.024)	-0.061* (0.032)	-0.080** (0.033)	-0.059* (0.031)	-0.117*** (0.038)	-0.141*** (0.040)
Firm x Day FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.475	0.472	0.500	0.486	0.492	0.513
N	5379	5350	5316	4267	4238	4204

(b) Gross positions						
	Gross securities positions			Gross combined positions		
	(1) h = 1	(2) h = 5	(3) h = 10	(4) h = 1	(5) h = 5	(6) h = 10
log(Limit Usage)	-0.009** (0.004)	-0.024*** (0.006)	-0.034*** (0.008)	-0.009*** (0.003)	-0.019*** (0.006)	-0.023*** (0.007)
Firm x Day FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.496	0.500	0.500	0.506	0.498	0.503
N	5379	5350	5316	4267	4238	4204

Table 4 shows the effect of log(Limit Usage) on the log change in net positions (Panel a) and on the log change in gross positions (Panel b).

Table 4: Firm-by-day fixed effects

Table 6
Instrumental variable analysis with limit changes at non-Treasury desks.

	Net securities positions			Gross securities positions		
	(1) h = 1	(2) h = 5	(3) h = 10	(4) h = 1	(5) h = 5	(6) h = 10
fitted log(Limit Usage)	-0.134* (0.073)	-0.256*** (0.095)	-0.245** (0.103)	-0.006 (0.013)	-0.069*** (0.022)	-0.077*** (0.026)
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
N	16816	16764	16699	16816	16764	16699

Table 6 shows the second stage results of instrumental variable regressions using $\Delta(\text{VaR Limit})^{\text{inc}}$ and $\Delta(\text{VaR Limit})^{\text{dec}}$ to estimate the effect of log(Limit Usage) on the log change in net securities positions (Columns 1 through 3) and on the log change in gross securities positions (Columns 4 through 6). $\Delta(\text{VaR Limit})^{\text{inc}}$ and $\Delta(\text{VaR Limit})^{\text{dec}}$ is computed as the percent change from day $t-2$ to $t-1$ of the sum of VaR limit size across all non-Treasury desks at the same firm that increased and decreased their VaR limit sizes between day $t-2$ to $t-1$, respectively. Limit Usage is defined as the ratio of VaR relative to VaR limit from the day before. Regression results are shown for positions change over $h = 1, 5, \text{ and } 10$ business days. The sample period is January 1, 2015 to March 31, 2023. All regressions include time fixed effects. Standard

Table 6: Instrumental-variable analysis using non-Treasury desk limit changes

Read:

- Table 4 adds firm-by-day fixed effects and still finds negative effects of limit usage on positions.
- The combined net position coefficients are -0.059, -0.117, and -0.141 at horizons 1, 5, and 10 days.
- Table 5 shows that newer assets are affected first: for assets younger than 30 days, coefficients are -0.010, -0.034, and -0.047 at horizons 1, 5, and 10.
- For older assets, the effects are smaller at one day but become significant over longer horizons.
- Table 6 shows IV coefficients on fitted log limit usage that are larger in magnitude than the baseline for net positions, especially over one- and two-week horizons.

Economic message. The results remain when comparing desks within the same firm on the same day. Dealers adjust newer inventory first, consistent with active inventory management rather than mechanical shrinking.

Table 5
Effect of VaR constraints on dealer long (assets) and short (liabilities) positions of different vintages.

(a) h = 1				
	(1) Assets (≤ 30 day)	(2) Assets (> 30 day)	(3) Liab (≤ 30 day)	(4) Liab (> 30 day)
log(Limit Usage)	-0.010*** (0.003)	-0.002 (0.002)	-0.008** (0.003)	-0.003 (0.003)
R ²	0.152	0.242	0.146	0.208
N	11 428	11 426	11 434	11 323
chi p-value	0.011		0.204	

(b) h = 5				
	(1) Assets (≤ 30 day)	(2) Assets (> 30 day)	(3) Liab (≤ 30 day)	(4) Liab (> 30 day)
log(Limit Usage)	-0.034*** (0.005)	-0.018*** (0.005)	-0.013** (0.006)	-0.017** (0.007)
R ²	0.168	0.150	0.150	0.165
N	11 376	11 374	11 382	11 260
chi p-value	0.016		0.711	

(c) h = 10				
	(1) Assets (≤ 30 day)	(2) Assets (> 30 day)	(3) Liab (≤ 30 day)	(4) Liab (> 30 day)
log(Limit Usage)	-0.047*** (0.006)	-0.029*** (0.008)	-0.029*** (0.007)	-0.020** (0.010)
R ²	0.176	0.151	0.146	0.160
N	11 360	11 358	11 316	11 189

Table 5: Long and short positions of different vintages

5.5 Tables 7-8: regulatory constraints and credit-line drawdowns

Table 7
Regulatory constraints and dealer positions.

(a) Full sample period						
	Net securities positions			Gross securities positions		
	(1) h = 1	(2) h = 5	(3) h = 10	(4) h = 1	(5) h = 5	(6) h = 10
log(Limit Usage)	-0.033 (0.022)	-0.070*** (0.027)	-0.071** (0.031)	-0.011** (0.005)	-0.033*** (0.007)	-0.043*** (0.008)
lagged SLR dist	-0.000 (0.014)	-0.005 (0.018)	-0.019 (0.019)	-0.001 (0.005)	-0.000 (0.006)	-0.009 (0.008)
log(Limit Usage) xlagged SLR dist	-0.002 (0.007)	0.006 (0.009)	-0.000 (0.010)	0.001 (0.002)	0.002 (0.003)	0.002 (0.003)
Desk FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.122	0.116	0.115	0.127	0.153	0.152
N	10 947	10 907	10 857	10 947	10 907	10 857

(b) 2020						
	Net securities positions			Gross securities positions		
	(1) h = 1	(2) h = 5	(3) h = 10	(4) h = 1	(5) h = 5	(6) h = 10
log(Limit Usage)	-0.063 (0.052)	-0.139** (0.070)	-0.120 (0.076)	-0.014* (0.008)	-0.035** (0.014)	-0.046*** (0.017)
lagged SLR dist	-0.003 (0.091)	-0.128 (0.122)	-0.117 (0.135)	-0.016 (0.016)	-0.036 (0.027)	-0.071** (0.031)
log(Limit Usage) xlagged SLR dist	0.001 (0.015)	0.002 (0.022)	-0.008 (0.024)	-0.003 (0.002)	-0.006 (0.004)	-0.005 (0.005)
Desk FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.122	0.116	0.115	0.127	0.153	0.152
N	10 947	10 907	10 857	10 947	10 907	10 857

Table 7: SLR distance and dealer positions

Table 8
Credit line drawdowns and dealer positions.

(a) Full sample period						
	Net securities positions			Gross securities positions		
	(1) h = 1	(2) h = 5	(3) h = 10	(4) h = 1	(5) h = 5	(6) h = 10
log(Limit Usage)	-0.032** (0.013)	-0.038** (0.018)	-0.055*** (0.020)	-0.007*** (0.003)	-0.022*** (0.004)	-0.029*** (0.005)
lagged SLR dist	0.000 (0.013)	-0.011 (0.017)	-0.014 (0.018)	-0.001 (0.004)	0.000 (0.006)	-0.008 (0.008)
drawdown	-0.199 (0.732)	-0.487 (0.361)	0.153 (0.238)	-0.536* (0.284)	-0.103 (0.154)	-0.101 (0.164)
lagged SLR dist xdrawdown	0.146 (0.437)	0.256 (0.204)	-0.142 (0.129)	0.304* (0.181)	0.030 (0.087)	0.044 (0.055)
Desk FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.163	0.161	0.168	0.171	0.186	0.193
N	8375	8339	8299	8375	8339	8299

(b) 2020						
	Net securities positions			Gross securities positions		
	(1) h = 1	(2) h = 5	(3) h = 10	(4) h = 1	(5) h = 5	(6) h = 10
log(Limit Usage)	-0.071** (0.028)	-0.103*** (0.039)	-0.151*** (0.040)	-0.011* (0.006)	-0.027*** (0.009)	-0.044*** (0.011)
lagged SLR dist	0.021 (0.078)	0.071 (0.113)	0.131 (0.113)	-0.002 (0.017)	-0.014 (0.026)	-0.028 (0.029)
drawdown	-1.179 (1.552)	-0.697 (0.531)	0.467 (0.306)	0.139 (0.275)	0.144 (0.132)	0.044 (0.094)
lagged SLR dist xdrawdown	0.916 (1.083)	0.440 (0.316)	-0.207 (0.165)	-0.045 (0.147)	-0.040 (0.063)	0.002 (0.042)

Table 8: Credit-line drawdowns and dealer positions

Read:

- Table 7 controls for SLR distance and the interaction between SLR distance and VaR limit usage.
- In most specifications, VaR limit usage remains negative and significant, while SLR terms are generally weak.
- In the 2020 subsample, the VaR effects on net and gross securities positions remain economically meaningful.
- Table 8 adds credit-line drawdowns as a higher-frequency proxy for balance-sheet pressure.
- Drawdowns do not overturn the VaR effect; in both full-sample and 2020 specifications, log limit usage remains negative in most columns.

Economic message. Internal risk limits are not just a disguised SLR effect. Size-based regulation may matter in some settings, but daily Treasury intermediation appears more directly constrained by VaR limits.

5.6 Tables 9-11: robustness and other risk limits

Table 9
Robustness.

(a) Long positions						
	Long securities positions			Long combined positions		
	(1) h = 1	(2) h = 5	(3) h = 10	(4) h = 1	(5) h = 5	(6) h = 10
log(Limit Usage)	-0.008*** (0.002)	-0.025*** (0.003)	-0.034*** (0.004)	-0.007*** (0.002)	-0.020*** (0.003)	-0.020*** (0.004)
R ²	0.134	0.153	0.161	0.133	0.143	0.154
N	16 838	16 786	16 721	12 045	11 986	11 915

(b) Short positions						
	Short securities positions			Short combined positions		
	(1) h = 1	(2) h = 5	(3) h = 10	(4) h = 1	(5) h = 5	(6) h = 10
log(Limit Usage)	-0.005* (0.003)	-0.013*** (0.004)	-0.023*** (0.005)	-0.005*** (0.001)	-0.009*** (0.003)	-0.015*** (0.003)
R ²	0.131	0.140	0.136	0.135	0.142	0.144
N	16 825	16 773	16 706	12 047	11 989	11 921

(c) Net positions - all Treasury market-making desks						
	Net securities positions			Net combined positions		
	(1) h = 1	(2) h = 5	(3) h = 10	(4) h = 1	(5) h = 5	(6) h = 10
log(Limit Usage)	-0.020*** (0.008)	-0.045*** (0.012)	-0.063*** (0.012)	-0.025** (0.011)	-0.053*** (0.014)	-0.075*** (0.016)
R ²	0.062	0.063	0.069	0.066	0.064	0.067
N	32 560	32 438	32 292	23 471	23 349	23 201

Table 9 shows the effect of log(Limit Usage) on the log change in long positions (Panel a) and on the log change in short positions (Panel b). Panel (c) shows the effect of log(Limit Usage) on the log change in net securities positions (columns 1 through 3) and on the log change in net combined positions (columns 4 through 6) for all UST market-making desks. Net positions are the absolute value of the difference between long and short positions. Securities positions include only cash assets. Combined positions include cash and derivative assets. Limit Usage is defined as the ratio of VaR relative to VaR limit from the day before. Regression results are shown for positions change over h = 1, 5, and 10 business days. The sample period is January 1, 2015 to March 31, 2023 for securities positions (columns 1 through 3) and January 1, 2015

Table 9: Robustness: long, short, and broader desk sample

Table 10
Robustness - limit usage rank transformation.

(a) Net securities positions								
	h = 1			h = 5			h = 10	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Limit Usage (std)	-0.010** (0.005)			-0.016** (0.006)			-0.023*** (0.007)	
Limit Usage pre rank		-0.044** (0.018)			-0.073*** (0.024)			-0.093*** (0.026)
Limit Usage q2			-0.007 (0.015)			-0.041** (0.020)		-0.064*** (0.022)
Limit Usage q3			-0.011 (0.016)			-0.028 (0.020)		-0.032 (0.022)
Limit Usage q4			-0.033** (0.015)			-0.059*** (0.019)		-0.078*** (0.021)
R ²	0.129	0.131	0.128	0.124	0.125	0.124	0.131	0.131
N	16 649	16 469	16 832	16 507	16 417	16 781	16 532	16 713

(b) Net combined positions								
	h = 1			h = 5			h = 10	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Limit Usage (std)	-0.007 (0.007)			-0.017** (0.009)			-0.038*** (0.010)	
Limit Usage pre rank		-0.046 (0.040)			-0.099*** (0.037)			-0.153*** (0.039)
Limit Usage q2			-0.028 (0.023)			-0.073** (0.030)		-0.078** (0.032)
Limit Usage q3			-0.019 (0.024)			-0.085*** (0.030)		-0.079** (0.032)
Limit Usage q4			-0.039 (0.024)			-0.062** (0.030)		-0.108*** (0.032)
R ²	0.131	0.133	0.131	0.125	0.127	0.125	0.143	0.145
N	11 853	11 673	11 882	11 794	11 614	11 823	11 723	11 543

Table 10: Alternative limit-usage rank transformations

Table 11

Interest rate risk limits.

(a) Net securities positions						
	h = 1		h = 5		h = 10	
	(1)	(2)	(3)	(4)	(5)	(6)
log(DV01 Limit Usage)	-0.007** (0.003)	-0.006** (0.003)	-0.013*** (0.004)	-0.011*** (0.004)	-0.015*** (0.004)	-0.012*** (0.004)
log(Limit Usage)		-0.029*** (0.011)		-0.054*** (0.014)		-0.066*** (0.016)
R^2	0.129	0.129	0.125	0.126	0.132	0.133
N	16 838	16 838	16 786	16 786	16 721	16 721
(b) Net combined positions						
	h = 1		h = 5		h = 10	
	(1)	(2)	(3)	(4)	(5)	(6)
log(DV01 Limit Usage)	-0.025** (0.010)	-0.019* (0.010)	-0.055*** (0.013)	-0.045*** (0.013)	-0.061*** (0.014)	-0.047*** (0.014)
log(Limit Usage)		-0.038** (0.017)		-0.073*** (0.022)		-0.097*** (0.023)
R^2	0.131	0.131	0.127	0.128	0.144	0.145
N	12 045	12 045	11 986	11 986	11 915	11 915
(c) Gross securities positions						
	h = 1		h = 5		h = 10	
	(1)	(2)	(3)	(4)	(5)	(6)
log(DV01 Limit Usage)	-0.001 (0.000)	-0.000 (0.000)	-0.001 (0.001)	-0.000 (0.001)	-0.002* (0.001)	-0.001 (0.001)
log(Limit Usage)		-0.006*** (0.002)		-0.020*** (0.003)		-0.030*** (0.003)
R^2	0.131	0.132	0.151	0.153	0.150	0.154
N	16 838	16 838	16 786	16 786	16 721	16 721
(d) Gross combined positions						
	h = 1		h = 5		h = 10	
	(1)	(2)	(3)	(4)	(5)	(6)
log(DV01 Limit Usage)	-0.003*** (0.001)	-0.002** (0.001)	-0.005*** (0.001)	-0.003* (0.002)	-0.005*** (0.002)	-0.002 (0.002)
log(Limit Usage)		-0.006*** (0.001)		-0.016*** (0.002)		-0.023*** (0.003)
R^2	0.135	0.137	0.142	0.146	0.149	0.154
N	12 045	12 045	11 986	11 986	11 915	11 915

Table 11: Interest-rate risk limits**Read:**

- Table 9 shows that both long and short positions decline with higher limit usage.
- The expanded sample of all Treasury-related desks also shows negative and significant effects.
- Table 10 shows that standardized usage, percentile ranks, and quartile indicators all produce negative effects.
- The quartile pattern is especially informative: desks in the highest usage quartile reduce positions more strongly.
- Table 11 shows that DV01 limits also matter, but VaR limit usage remains significant after controlling for DV01 limit usage.

Economic message. The main finding is robust across outcome definitions, sample definitions, and transformations of the limit-usage variable. VaR limits capture a broader portfolio risk constraint than interest-rate sensitivity alone.

5.7 Figure 5, Figure 6, and Table 12: selling to the Fed

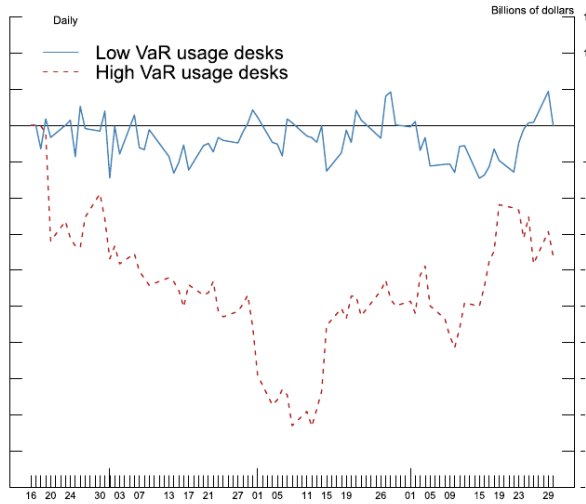


Figure 5: Inventory change since March 16, 2020 by VaR usage

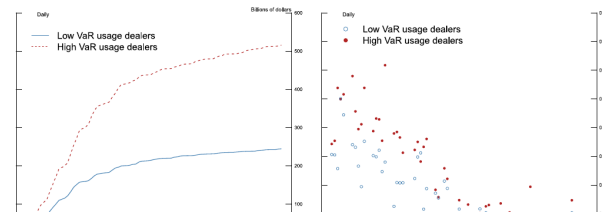


Figure 6: Treasury sales to the Fed grouped by VaR usage

Table 12
Dealer sales to SOMA.

Panel A						
	Probability of sale			Log amount of sale		
	(1)	(2)	(3)	(4)	(5)	(6)
$\log(\text{Limit Usage})_{t-1}$	0.167*** (0.022)	0.117*** (0.039)		0.321*** (0.113)	0.751*** (0.244)	
March = 0 × $\log(\text{Limit Usage})_{t-1}$			0.030 (0.038)			0.068 (0.190)
March = 1 × $\log(\text{Limit Usage})_{t-1}$			0.127*** (0.047)			0.373* (0.215)
Dealer FE		Yes	Yes		Yes	Yes
Time FE	Yes		Yes	Yes		Yes
R^2	0.244	0.451	0.616	0.499	0.331	0.750
N	619	619	619	520	520	520
Panel B						
	Sales to SOMA/tot sales to cust.			Sales to SOMA/tot buys from cust.		
	(1)	(2)	(3)	(4)	(5)	(6)
$\log(\text{Limit Usage})_{t-1}$	0.058*** (0.010)	0.100*** (0.023)		0.104*** (0.016)	0.140*** (0.036)	
March = 0 × $\log(\text{Limit Usage})_{t-1}$			0.027 (0.022)			0.041 (0.036)
March = 1 × $\log(\text{Limit Usage})_{t-1}$			0.059** (0.026)			0.085** (0.041)
Dealer FE		Yes	Yes		Yes	Yes
Time FE	Yes		Yes	Yes		Yes
R^2	0.370	0.244	0.555	0.306	0.210	0.451

Table 12: Dealer sales to SOMA

Read:

- Figure 5 shows high-VaR-usage desks reducing net securities positions much more sharply after March 16, 2020.
- Figure 6 shows that high-VaR-usage dealers sold more Treasuries to the Fed, both in total dollar volume and relative to customer-sale volume.

- Table 12 shows that prior-day log limit usage predicts a higher probability of selling to SOMA and a larger log amount of sale.
- A 100 percent increase in dealer VaR usage implies dealers are 16.7 percent more likely to sell Treasuries to the Fed in the simplest probability specification.
- SOMA sales as a fraction of client intermediation also rise with limit usage.

Economic message. Fed purchases were used disproportionately by the dealers most constrained by VaR limits. This supports the idea that SOMA operations relieved risk constraints by taking securities off dealer books.

5.8 Tables 13-14: auction prices and client liquidity provision

to the Fed SOMA portfolio. In Panel B, the dependent variable for columns (1) through (3) is the ratio of dealer sales to SOMA relative to total dealer sales to customers that day, and the dependent variable for columns (4) through (6) is the ratio of dealer sales to SOMA relative to total dealer buys from customers that day. Trading volumes in the denominators are restricted to CUSIPs with SOMA auctions on that same day. Data are aggregated to the dealer-day level. Limit Usage is defined as the ratio of VaR relative to VaR limit from the day before. Standard errors are robust.

Table 13
Dealer SOMA offer-aggressiveness and VaR limit usage.

	$dp = \text{Avg Price in Auction} - \text{Price}_{\text{dealer sell}}$					
	Full sample (1)	(2)	(3)	Sample of Level 1 and 2 match (4)	(5)	(6)
$\log(\text{Limit Usage})_{t-1}$	0.008*** (0.002)			0.009*** (0.002)		
March = 0						
× $\log(\text{Limit Usage})_{t-1}$		0.006*** (0.001)			0.006*** (0.001)	
March = 1						
× $\log(\text{Limit Usage})_{t-1}$		0.009** (0.004)			0.012*** (0.004)	
Bond Life < 5Yrs			0.001 (0.001)			0.001 (0.001)
× $\log(\text{Limit Usage})_{t-1}$			0.007* (0.004)			0.007* (0.004)
Bond Life ∈ [5,10]Yrs			0.007* (0.004)			0.007* (0.004)
× $\log(\text{Limit Usage})_{t-1}$			0.021*** (0.008)			0.026*** (0.007)
Bond Life ≥ 10Yrs			0.008 (0.008)			0.007 (0.007)
× $\log(\text{Limit Usage})_{t-1}$			-0.002 (0.002)			-0.001 (0.002)
constant	-0.002 (0.002)	-0.002 (0.002)	-0.002 (0.002)	-0.001 (0.002)	-0.001 (0.002)	-0.001 (0.002)
Bond FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.053	0.053	0.054	0.070	0.070	0.072
N	11,012	11,012	11,007	10,244	10,244	10,239

Table 13 shows the effect of $\log(\text{Limit Usage})$ on the aggressiveness of dealers' offers in SOMA auctions during the period from March 16,

Table 14
Dealer liquidity provision to clients and SOMA operations.

	Log (dealer buy from customers)					
	SOMA (1)	Not SOMA (2)	Total (3)	SOMA (4)	Not SOMA (5)	Total (6)
$\log(\text{Limit Usage})_{t-1}$	-0.146*** (0.055)	-0.124** (0.063)	-0.152*** (0.046)	-0.149*** (0.054)	-0.129** (0.061)	-0.156*** (0.044)
$\log(\text{Sell to SOMA})$	0.048*** (0.016)	0.032 (0.021)	0.032** (0.013)			
March = 0						
× $\log(\text{Sell to SOMA})$				0.038** (0.017)	0.015 (0.019)	0.017 (0.013)
March = 1				0.085*** (0.018)	0.096** (0.038)	0.085*** (0.015)
× $\log(\text{Sell to SOMA})$						
Dealer FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.884	0.851	0.908	0.885	0.853	0.910
N	510	510	510	510	510	510

Table 14 shows the effect of $\log(\text{Limit Usage})$ and the SOMA operations on dealers' ability to supply liquidity to clients by taking bonds off their hands. The sample period is from March 16, 2020 to June 11, 2020. Transaction-level data are aggregated at the dealer-day level for this analysis. The dependent variable in columns (1) and (4) is the Log volume of trades that are dealer buy from customers for bonds that are included in the list of bonds with non-zero SOMA auction volume that day. The dependent variable in columns (2) and (5) is the Log volume of trades that are dealer buy from customers for bonds that are not in the list of bonds with a SOMA auction that day. The dependent variable in columns (3) and (6) is the Log volume of trades that are dealer buy from customers for all bonds. Limit Usage is defined as the ratio of

Table 13: Dealer SOMA offer aggressiveness

Table 14: Dealer liquidity provision to clients and SOMA operations

Read:

- Table 13 shows that constrained dealers submit more aggressive offers in SOMA auctions: they are willing to sell at lower prices.
- The effect is stronger in March and for longer-maturity bonds, which contribute more to portfolio VaR.
- Table 14 shows that dealers with higher prior-day limit usage buy less from clients.
- After controlling for VaR usage, selling more to the Fed is associated with buying more from clients, especially in March 2020.
- The positive spillover appears for bonds on and off the SOMA auction list, indicating fungible balance-sheet relief.

Economic message. The Fed did not merely receive bonds that clients wanted to sell. By removing inventory risk, the Fed increased dealer capacity to provide liquidity to clients more broadly.

5.9 Tables 15-16: profitability and aggregate market liquidity

Table 15

Dealer profitability and VaR constraint.

(a) h = 1			
	(1) Total	(2) New	(3) Existing
log(Limit Usage)	0.109*** (0.018)	0.074*** (0.014)	0.025 (0.023)
R^2	0.213	0.153	0.178
N	16 667	16 665	16 665
(b) h = 5			
	(1) Total	(2) New	(3) Existing
log(Limit Usage)	0.118*** (0.015)	0.092*** (0.012)	0.029 (0.020)
R^2	0.228	0.162	0.180
N	16 772	16 771	16 771
(c) h = 10			
	(1) Total	(2) New	(3) Existing
log(Limit Usage)	0.130*** (0.015)	0.102*** (0.012)	0.031 (0.019)
R^2	0.232	0.174	0.180
N	16 772	16 771	16 771

Table 15: Dealer profitability and VaR constraints

Table 16

Treasury market illiquidity quantile regressions.

(a) Yield curve fitting error				
	(1) OLS	(2) 50th	(3) 75th	(4) 99th
MOVE	0.719*** (0.014)	0.646*** (0.020)	0.723*** (0.028)	1.418*** (0.089)
Limit Usage Index residual	0.253*** (0.014)	0.181*** (0.025)	0.235*** (0.031)	0.238*** (0.043)
R^2	0.583			
Pseudo R^2		0.274 2023	0.378 2023	0.502 2023
(b) 10 year Treasury on the run premia				
	(1) OLS	(2) 50th	(3) 75th	(4) 99th
MOVE	0.737*** (0.015)	0.839*** (0.021)	0.940*** (0.020)	1.027*** (0.087)
Limit Usage Index residual	0.085*** (0.015)	0.091*** (0.025)	0.090*** (0.016)	0.142** (0.067)
R^2	0.552			
Pseudo R^2		0.347 2026	0.368 2026	0.511 2026
N	2026	2026	2026	2026

Table 16: Treasury market illiquidity quantile regressions

Read:

- Table 15 shows that future PnL per unit of VaR rises with limit usage, especially for PnL from new trades.
- The effect is statistically significant over one-day, one-week, and two-week horizons.
- This means constrained desks require more compensation to take on additional risk.
- Table 16 shows that the aggregate dealer VaR constraint index predicts Treasury market illiquidity even after controlling for MOVE volatility.
- The effect is stronger in higher illiquidity quantiles, especially the 99th percentile.

Economic message. Desk-level constraints become market-level liquidity risk. When dealers are collectively near their VaR limits, Treasury market liquidity is worse, especially in the tail.

6 Short summary

- The paper studies why dealers can behave as risk averse in the Treasury market.
- The main mechanism is internal trading-desk VaR limits.
- VaR limits are persistent, costly to breach, and sensitive to market volatility.
- The paper uses confidential Volcker rule supervisory data at the trading-desk level.
- Treasury desks reduce positions when their VaR limit usage rises.
- The response is stronger for net positions than gross positions, which supports a risk-based interpretation.
- Results survive firm-by-day fixed effects, IV analysis, regulatory controls, credit-line drawdowns, and alternative limit-usage measures.

- SLR distance does not explain the high-frequency changes in Treasury desk positions as well as VaR limit usage does.
- During March 2020, constrained dealers sold more Treasuries to the Fed and accepted lower prices.
- Fed SOMA purchases relieved constraints and allowed dealers to provide more liquidity to clients.
- Constrained dealers required higher compensation per unit of VaR for new trades.
- An aggregate VaR-constraint index is related to Treasury market illiquidity, especially in the tail of the illiquidity distribution.

7 Conclusion

7.1 Core conclusion

Li, Petrasek, and Tian show that internal trading-desk risk limits materially constrain Treasury dealer intermediation. The effect is particularly important in stress episodes, when volatility raises VaR and pushes desks toward their limits. Dealers close to their limits reduce risk, charge higher compensation, sell more to the Fed, and provide less liquidity to clients.

7.2 Economic intuition

The intuition is that VaR limits convert volatility into a binding capacity constraint. A dealer does not need to be intrinsically risk averse. If taking more inventory increases the probability of breaching a desk-level limit, then additional risk has a shadow cost. In March 2020, this shadow cost rose sharply. Fed purchases helped because they removed risk from dealer books, lowering the shadow cost and freeing capacity for client trades.

7.3 Takeaway

The paper provides a micro-foundation for the volatility-liquidity link in Treasury markets. A volatility shock does not only change prices directly; it raises measured dealer risk, tightens internal constraints, and reduces dealer balance-sheet capacity. Policy interventions that remove risky inventory from dealer books can therefore be more effective during crises than interventions aimed only at regulatory balance-sheet relief.